Assignment Set 2

CS-423 (Artificial Intelligence)

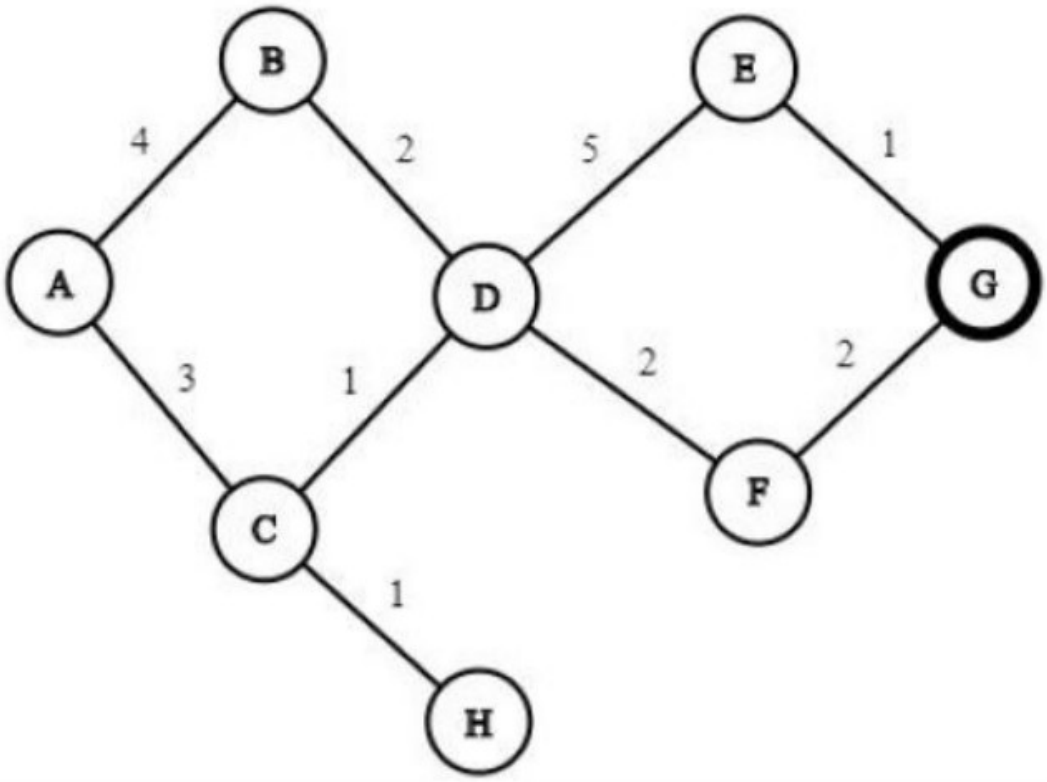
PART -III (Total Marks 1*10=10)

1) Suppose we do Iterative Deepening Search (IDS) but instead of increasing depths as 1,2,3,4… we increase it as 1,2,4,8,16… i.e. double the depth instead of increasing it by

1. This modified version of IDS is

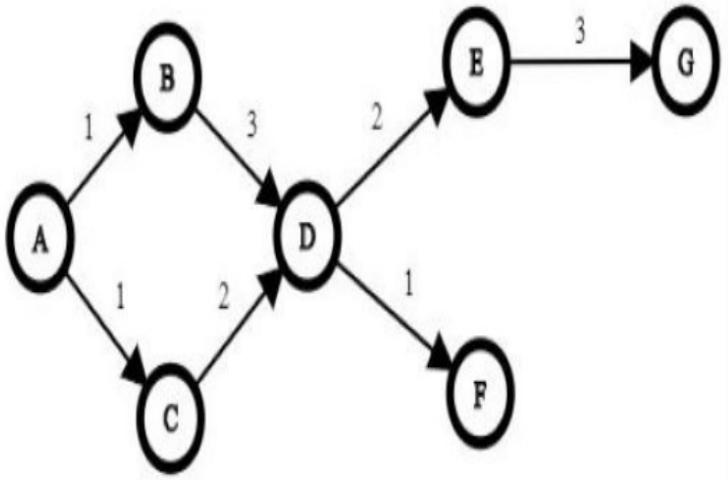
- [ ] Optimal but not complete
- [x] Complete but not optimal
- [ ] More efficient than standard IDS in the worst case
- [x] Less efficient than standard IDS in the worst case

3) Consider the following graph in which we are searching from start state A to goal state G. The number over each edge is the transition cost. Find the path to the goal found by Depth First Search with full duplicate detection, which explores children in lexicographical order. (Write answer as a string with no spaces. For example, if the order of exploration is A followed by B followed by C followed by D then write ABCD)



ABDEG

4) Consider the following directed graph in which we are searching from start state A to goal state G. The number over each edge is the transition cost. Find the sequence of nodes explored (i.e., the sequence of removals from its frontier) by Uniform Cost Search, with no duplicate detection. Assume that, in case of ties, the search procedure uses a lexicographical order for tie-breaking. Output the order of nodes explored. (Write answer as a string with no spaces. For example, if the order of exploration is A followed by B followed by C followed by D then write ABCD)



ABCDDFEFEG

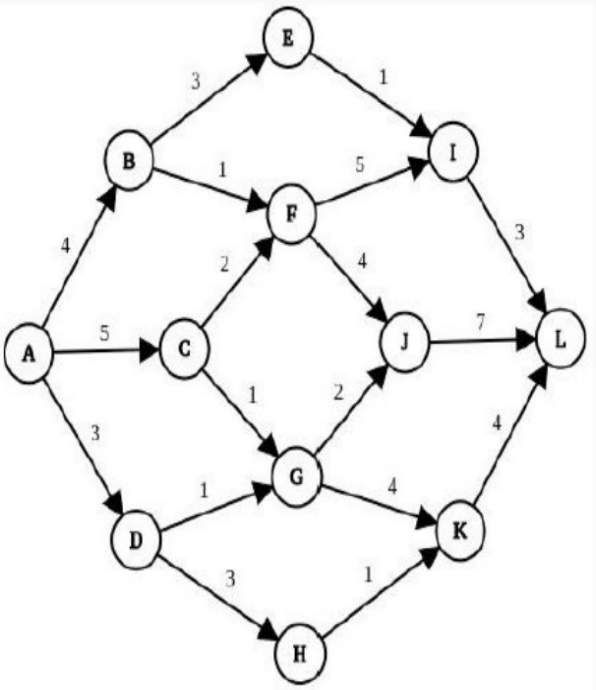
5) Which data structure is the most suited for implementing Uniform Cost Search?

- ○ Queue
- ○ Skip Lists
- ✓ Min Heap
- ○ Max Heap

6) Which of the following uninformed search method(s) can be used to compute a solution to the N-Queens problem?

- ☐ Bi-Directional Search
- ☐ Beam Search
- ☑ Iterative Deepening Search
- ☑ Breadth-First Search

7) Consider the following graph in which we are searching from start state A to goal state L. The number over each edge is the transition cost. Find the path to the goal found by Beam Search with a beam size of 2. The best partial solutions are the ones with the least cost. (Write the answer as a string with no spaces. For example, if the order of exploration is A followed by B followed by C followed by D then write ABCD). Note: Conflicts are resolved by choosing a lexicographically shorter path if both have the same cost.

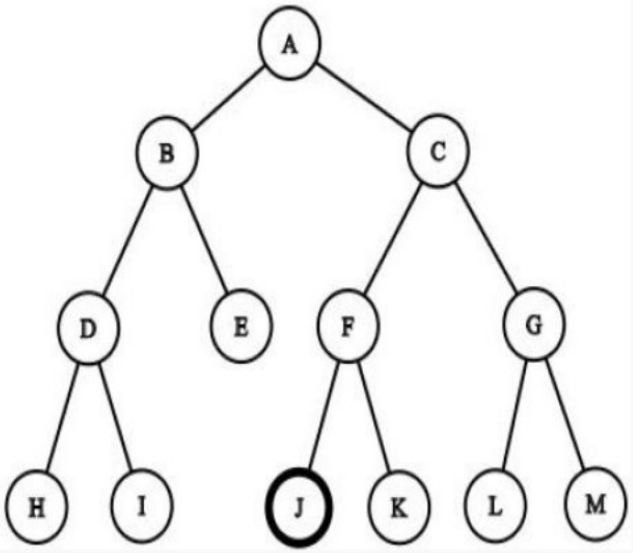


ADGKL

8) Consider a full binary tree (Every internal node has degree 2 and there are 2^n leaves) of height n. Enumerate the nodes in the level at height (n - 1) **1 poin** from 1 to 2^{n-1} (left to right) and let the goal state X be the $(2^{n-2} + 1)$ th node at that level. What is the absolute difference between the number of nodes explored by depth-first search and breadth-first search algorithms starting at the root? Assume n > 1 and left-to-right exploration.

- ✓ $2^{n-2} + n - 1$
- 2^{n-1}
- $2n - 2$
- $2^n - 2n + 2$

9) Consider the following graph in which we are searching from start state A to the goal state J. What is the number of nodes generated by Iterative Deepening Search? Assume that nodes are explored in right-to-left order and all edge costs are 1. Note: Exploration of the goal state J should be added to the count.



19

10) Consider the depth-limited search algorithm with depth limit f. If the only goal state exists at depth d then which of the following statement(s) is/are correct?

- [ ] If f > d then it is complete but not optimal.
- [x] If f > d then it is not complete and not optimal.
- [ ] If f < d then it is complete but not optimal.
- [x] If f < d then it is not complete and not optimal.